

```
import pandas as pd
df=pd.read_csv("test.csv")
df.head()
```

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S

Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 11 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  418 non-null    int64
1   Pclass       418 non-null    int64
2   Name         418 non-null    object
3   Sex          418 non-null    object
4   Age          332 non-null    float64
5   SibSp        418 non-null    int64
6   Parch        418 non-null    int64
7   Ticket       418 non-null    object
8   Fare         417 non-null    float64
9   Cabin        91 non-null     object
10  Embarked     418 non-null    object
dtypes: float64(2), int64(4), object(5)
memory usage: 36.1+ KB
```

```
df.describe()
```

	PassengerId	Pclass	Age	SibSp	Parch	Fare
count	418.000000	418.000000	332.000000	418.000000	418.000000	417.000000
mean	1100.500000	2.265550	30.272590	0.447368	0.392344	35.627188
std	120.810458	0.841838	14.181209	0.896760	0.981429	55.907576
min	892.000000	1.000000	0.170000	0.000000	0.000000	0.000000
25%	996.250000	1.000000	21.000000	0.000000	0.000000	7.895800
50%	1100.500000	3.000000	27.000000	0.000000	0.000000	14.454200
75%	1204.750000	3.000000	39.000000	1.000000	0.000000	31.500000
max	1309.000000	3.000000	76.000000	8.000000	9.000000	512.329200

```
df.columns
```

```
Index(['PassengerId', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch',
      'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
```

Generate

create a dataframe with 2 columns and 10 rows

Close

```
df.isnull().sum()
```

```

0
PassengerId    0
Pclass         0
Name           0
Sex            0
Age           86

```

```
df['Age'].fillna(df['Age'].median(),inplace=True)
```

```

<ipython-input-8-e0c0624688bb>:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value, inplace=True)
Cabin         327
df['Age'].fillna(df['Age'].median(),inplace=True)
Embarked      0

```

```
df['Fare'].fillna(df['Fare'].median(),inplace=True)
```

```

<ipython-input-10-ca3880b56554>:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value, inplace=True)
df['Fare'].fillna(df['Fare'].median(),inplace=True)

```

```
df.drop('Cabin',axis=1,inplace=True)
```

```

if df['Embarked'].isnull().sum() > 0:
    df['Embarked'].fillna(df['Embarked'].mode()[0],inplace=True)

```

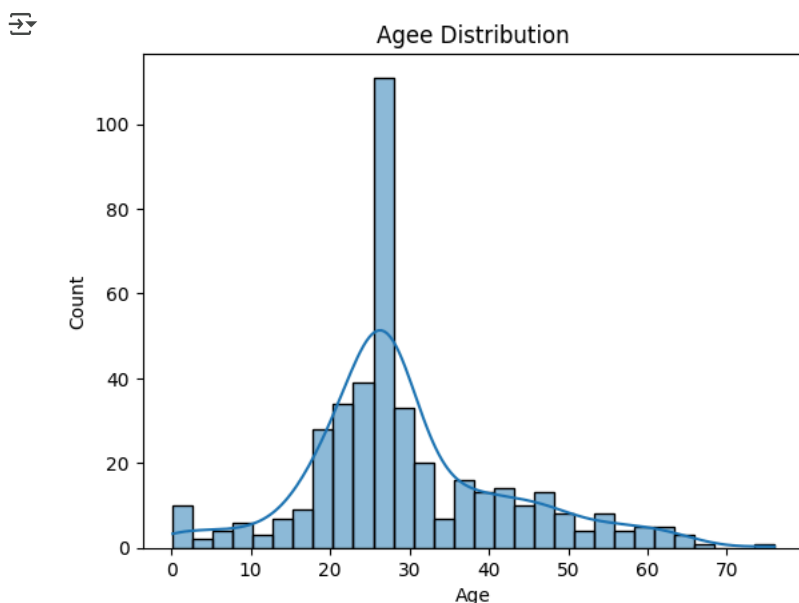
```
df['Sex']=df['Sex'].map({'male':0,'female':1})
```

```
df['Embarked']=df['Embarked'].map({'S':0,'C':1,'Q':2})
```

```

import matplotlib.pyplot as plt
import seaborn as sns
sns.histplot(df['Age'],bins=30,kde=True)
plt.title("Age Distribution")
plt.show()

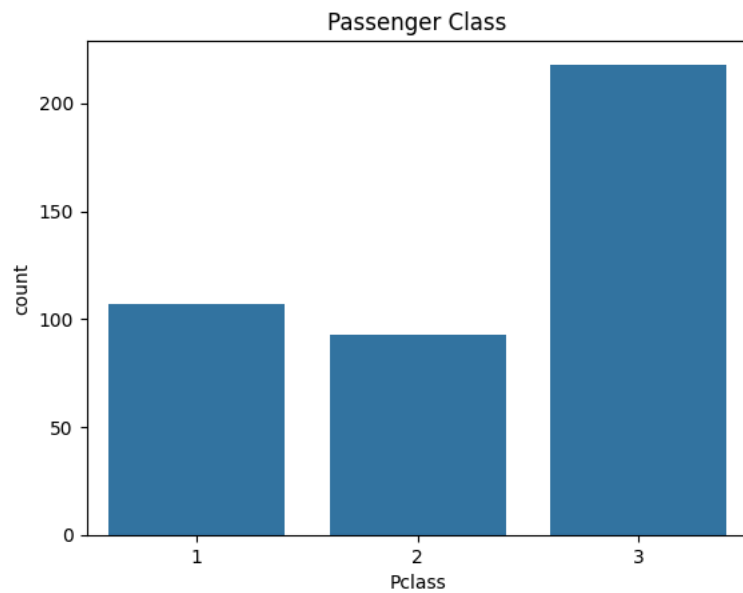
```



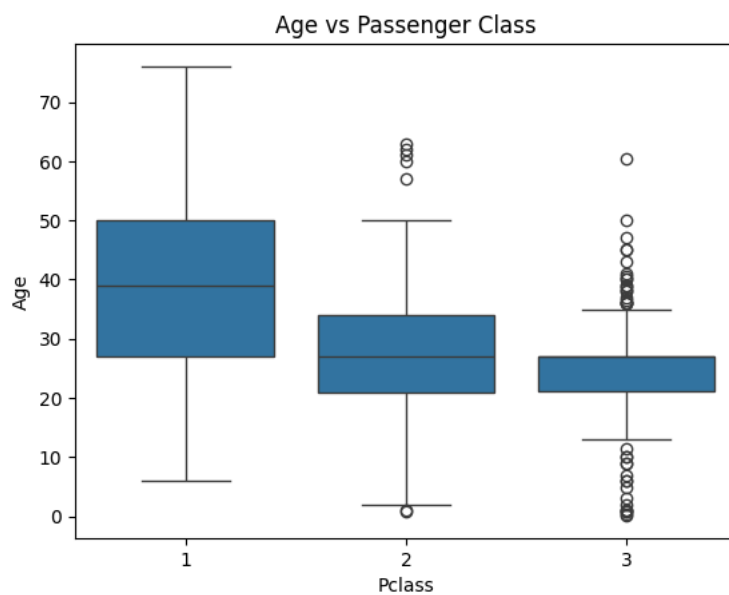
```

sns.countplot(x='Pclass',data=df)
plt.title("Passenger Class")
plt.show()

```



```
sns.boxplot(x='Pclass',y='Age',data=df)
plt.title("Age vs Passenger Class")
plt.show()
```



```
sns.violinplot(x='Sex',y='Age',data=df)
plt.title("Age vs Sex")
plt.show()
```



Age vs Sex

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
```

```
# Assuming df is already loaded and preprocessed as in the previous cells
```

```
# Select only numeric columns for correlation calculation
df_numeric = df.select_dtypes(include=np.number)
```

```
plt.figure(figsize=(10,6))
# Calculate and plot the correlation heatmap using the numeric columns
sns.heatmap(df_numeric.corr(),annot=True,cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

